

Disprove:

Statements, open sentences

known
to be true

(Proposition, Theorem)

Truth

unknown

(Conjecture)

known

to be

false

(i) Pythagorean theorem

(i) All perfect numbers are even

(i) All primes are odd

(ii) Goldbach conjecture

(ii) $0 = 1$

(iii) Poincaré conjecture

(iii) $a^n + b^n = c^n$
 $a, b, c \in \mathbb{N}$

to prove or disprove conjecture

$n > 2$

$\downarrow$
Euler conjecture
(disproved)

(rice examples)

Three major techniques:

(i) direct proof

(ii) contrapositive

(iii) contradiction

say, we have a mathematical statement. P
we want to disprove it,

we need to show $\sim P$: true, P : False

$\neg P$: we can use any one of the above 3 mentioned methods

however, in many cases our statement consists of Quantifier, $\forall$: for every, $\exists$: there exists
we have a statement with $\forall$:

$$\forall x \in S, P(x)$$

to disprove it

$$\neg (\forall x \in S, P(x)) = \underline{\exists x \in S, \neg P(x)}$$

there exists at least one
 $x \in S, P(x)$ is false

disproving by
example.

conjecture:

$$\forall n \in \mathbb{Z}, \underline{f(n) = n^2 - n + 11 \text{ is prime}}$$

$P(n)$

$\forall P(n)$ is true

$$n \quad 0 \quad 1 \quad -1 \quad 2 \quad -2 \quad 3 \quad -3$$

$$f(n) \quad 11 \quad 11 \quad 13 \quad 13 \quad 15$$

→ not a prime

hence, $2 \in \mathbb{Z}$, $f(2)$ is not prime

Conjecture:

if P then Q

$$P \Rightarrow Q$$

disprove

we need to generate a case where P is true however, Q is false

$P \Rightarrow Q$: is false

if A, B, C are sets, $A - (B \cap C) = (A - B) \cap (A - C)$

P : a test case $\Rightarrow Q$

$$A = \{1, 2, 3\}, B = \{1, 2\}, C = \{2, 3\}$$

$$A - (B \cap C) = \{1, 2, 3\} - (\{1, 2\} \cap \{2, 3\})$$

$$= \{1, 3\}$$

$$(A - B) \cap (A - C) = (\{1, 2, 3\} - \{1, 2\}) \cap (\{1, 2, 3\} - \{2, 3\})$$

$$= \{3\} \cap \{1\} = \emptyset$$

$$\therefore A - (B \cap C) \neq (A - B) \cap (A - C)$$

$\neg Q$

if $n \in \mathbb{Z}$ $n^5 - n$ is even, then n is even,

P

Q

$n = 0$

$n^5 - n = 0$: even

n even

1

0 : even

1 : odd

2

hence $\exists i \in \mathbb{Z}, i^2 - 1 = 0$: even
however, i is odd

• if $a, b \in \mathbb{N}$, $a + b \leq ab$

a	b	$a + b$	ab	$a + b \leq ab$
1	1	2	1	F
2	1	3	2	F
2	2	4	4	F
3	2	5	6	T

• if $A \subseteq B$ are sets,

$$\underline{|A \cup B| = |A| + |B|}$$

$$A = \{1, 2, 3\} \quad B = \{1, 3\}$$

$$A \cup B = \{1, 2, 3\}$$

$$|A \cup B| = 3$$

$$|A| = 3 \quad |B| = 2 \quad \Rightarrow 3 + 2 = 5$$

so, for the above case the statement is
not true,

$$|A \cup B| \leq |A| + |B|$$

∴:

$$P \Rightarrow Q$$

Disproving Existence statements:

$$\exists x \in S, P(x) \quad \vee \quad (\exists x \in S, P(x))$$

$$= \forall x \in S, \neg P(x)$$

- There is a real x for which $x^4 \leq x \leq x^2$
disprove this with examples

$$x = \frac{1}{2} \quad x^4 \leq x : \text{True} \quad \text{but} \quad x \leq x^2 : \text{False}$$

$$x = 2 \quad x \leq x^2 : \text{True} \quad \text{but} \quad x^4 \leq x : \text{False}$$

Disproving with examples becomes a bit cumbersome, hence, $\exists x \in \mathbb{R}$

$$\neg (\exists x \in \mathbb{R}, x^4 \leq x \leq x^2) = \forall x \in \mathbb{R}, \neg (x^4 \leq x \leq x^2)$$

↓
contradiction

Disprove by contradiction:

to disprove P with contradiction,

we start with P : True

deduce contradiction

we assume, $\exists x \in \mathbb{R}, x^4 \leq x \leq x^2$

since, $x \neq 0$, we can divide x on each part
 $x > 0$

since
 $x \in \mathbb{R}$
 $x^4 > 0$
 $0 < x^4 \leq x$
 $0 < x$

$$x^0 < 1 < x$$

$$0 < (x-1)$$

$$\rightarrow x^3 - 1 < 0 < x - 1$$

$$x > 1$$

$$\rightarrow (x-1)(x^2 + x + 1) < 0 < (x-1)$$

dividing with $(x-1)$

$$\rightarrow (x^2 + x + 1) < 0 < 1$$

we know, $x > 1$, $x > 0$

$$\therefore x^2 + x + 1 > 0$$

$$\rightarrow x^2 + x + 1 < 0$$

we have a contradiction

• $\forall a, b \in \mathbb{Z}$ for which $42a + 7b = 1$
disprove by contradiction,

$$\forall a, b \in \mathbb{Z} \quad 42a + 7b = 1 \text{ is true}$$

$$\text{but, } 42a + 7b = 7(6a + b)$$

$$\text{Now, } \forall a, b \in \mathbb{Z}, \quad 6a + b \in \mathbb{Z}$$

$$\therefore 7 \cdot (6a + b) \in 7 \cdot \mathbb{Z}$$

$$\text{since } 1 \notin 7 \cdot \mathbb{Z}$$

$\therefore$ we arrive at a contradiction

$$\therefore \text{Hence, } \forall a, b \in \mathbb{Z}, \quad 42a + 6b \neq 1$$

$$42a + 6b = 1$$

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